MODULE 2 · THE AUTOMATION LADDER

Scaling RPA + the Center of Excellence

Ten bots is a demo. A thousand is an operation — and someone has to run it.

Friday, October 2, 2026 · 8:00–10:30 AM · 2 MTC Room 802

IE-GY 9113B — Systems Integration: From ERP to Agentic AI · Prof. Emanuele Cimica

Today: rebuild the spine, work Telefónica O2, then design your CoE

The note gave you the concept. Today you apply it to your own company and design the organisation that scales it.

1	**Activation**	**8:05**	Live, timed team retrieval — not a recap. Your answers are the entry ticket to the breakout.
2	**Reconnect + story**	**8:20**	From your RPA scan to the scaling cliff; then Telefónica O2 — the programme that actually scaled.
3	**The CoE, worked**	**8:34**	What breaks at scale, the operating model that answers it, and reading a vendor case critically.
4	**Apply — team build**	**9:02**	Three asks in one working block while I come round; then each team reports out — your CoE / operating model (Board Memo §4, slide S7).
5	**Prime Lab 2**	**10:08**	You'll build your Lab 2 first pass at home; class breaks it and adds oversight.
6	**Homework + wrap**	**10:18**	The CoE section · build your Make first pass · keep the board deck evergreen.

8:03

BEFORE WE START

20 bots: a triumph.
400 bots: a crisis nobody owns.

The pilot is the easy part. Today is what happens next — why programmes die on the way to scale, and the organisation that keeps them alive. We start now, together.

8:05

Q1 — Where are we on the ladder?

make agentic	AI agents (Module 3) → the agentic enterprise (Module 4)
make intelligent	Intelligent automation (Module 3)
automate	RPA — still Module 2. Last week you placed one bot; this week you run a hundred. ◀ **you are here**
integrate	The integration backbone — one single source of truth (Module 1)

One programme, three threads. Last week it was one robot, three threads. This week the same threads — Customer, Operations & Supply Chain, Cash — at the scale of a whole programme.

8:07

Q2 — Your pilot worked. Why do most programmes die at scale?

The demo

Two weeks, a happy sponsor.

Pick a high-volume, rules-based step, point a bot at it, and in weeks you have a working automation. The most reliable demo in enterprise tech — the play you learned in Week 4.

The operation

A hundred bots nobody owns.

The same surface-level fragility that made one bot cheap makes a thousand expensive to keep alive. Screens move, integrations drift, the plant buckles, and bots multiply with no owner.

Two-sided question. Why does a programme that works brilliantly small break as it grows — and what organisation, deliberately built, keeps it alive? That second half is your Board Memo §4.

8:09

Q3 — Name the four ways a programme breaks at scale

Think 30 s → confer with your team → **a team gets cold-called**

You met all four in the note. Two are engineering problems. Two are… you tell me.

1	2	3	4

Name them first — then we reveal. And once they're up, we vote on which one kills most programmes.

The four ways a programme breaks at scale

Selector instability

A screen moves; the script fails.

A vendor update renames a field — the bot stops, or runs on confidently.

Integration drift

The estate moves under the bots.

Dozens of systems, each patched on its own schedule; the surface drifts.

Infrastructure brittleness

The pilot plant can't run the programme.

What carried ten bots collapses under hundreds.

Bot sprawl / shadow IT

Bots nobody owns.

Easy to build, no sanctioned path — teams stand up their own, unmonitored.

Hands up — *which one kills most programmes? Vote for each as we point.*

Three are engineering; the fourth is organisation. Bot sprawl has no technical cause — and it is the one this week answers.

8:12

Q4 — What does a Center of Excellence do? Name its five functions

Think 30 s → confer with your team → **a team gets cold-called**

You named the disease — bot sprawl. Now the cure: five jobs one organisation owns.

1	2	3	4	5

Cover them as a team before we reveal. If you can name five jobs, you can design one for your company in the breakout.

A Center of Excellence — and its five functions

Governance & standards	Who may build, how, and how a bot is approved before it goes live.
Intake & review	A sanctioned front door for demand — the antidote to the shadow door.
Bot ownership & monitoring	Every bot has an owner and a watched pulse; the audit trail catches a silent error.
Enablement & training	Turns willing business people into capable, supervised builders.
Infrastructure & platform	Owns the plant and scales it in step with the programme.

The CoE governs; it does not personally build everything. It makes automation a governed capability, not a pile of scripts.

8:15

Q5 — Scarcity says hoard the experts. Scale says…?

Scarcity mindset

Hoard a few experts.

A central team builds one at a time. The queue is a lifetime of work; the ninety-ninth process never gets touched.

Scale mindset

Multiply the builders.

Capture the flood safely — a dam with guardrails: many supervised builders under the CoE's governance.

The dam, not the bottleneck. A CoE that only builds is a bottleneck with a nicer name — and a sanctioned way to build is the direct answer to shadow IT.

8:16

Q6 — “Rules-based” does not mean “no human needed.” Where does a human stay?

Think 30 s → confer with your team → **a team gets cold-called**

A bot follows a wrong rule with perfect confidence. So where do you keep a person — and why there?

Decide the ‘where’ as a team before we turn the page. This is one of your three study questions — and it seeds Board Memo §5.

Automate the routine; route the judgement — on purpose

The trap

A wrong rule, run with confidence.

A bot doesn't pause, doubt, or notice the thing a person would catch out of the corner of their eye.

The move

Route the exceptions on purpose.

Automate the routine; send edge cases, high-value transactions, and irreversible actions to a human checkpoint.

Accountability is relocated, not removed. If a bot is silently wrong for weeks — who is accountable? That's the week's ethics thread, and the CoE is where it lands.

The spine, in one line

A pilot proves the logic; it does not prove the plant that runs it at ten times the volume. Scaling is an operating-model decision before it is a technology one.

→ **What breaks —** selectors, integrations, infrastructure — and the organisational one, bot sprawl.

→ **The answer —** a Center of Excellence — governance, ownership, enablement — that makes automation a hard-to-copy capability.

→ **Your job today —** design that operating model for your company → Board Memo §4 (slide S7).

This week you design the organisation, not another bot. The deliverable is the operating model a board funds — the CoE / operating-model section of your memo.

8:19

Your ticket into the breakouts

IN YOUR TEAM, WRITE ONE LINE

The single biggest scaling risk on your RPA candidate — the stable-rules step from your §4 scan.

That line is your entry ticket to Breakout 1.

No line, no ticket. Breakout 1 starts from exactly what you write here.

You named where a bot belongs. Now: who runs a hundred of them?

From Week 4

Your RPA-candidate scan (§4) named the stable-rules steps worth a bot. One robot, three threads.

The turn

A pilot at three bots behaves nothing like a programme at three hundred — the fragility you met last week, multiplied.

This week

Design the operating model that scales it and keeps it alive — the CoE section that extends §4.

Last week: where a bot belongs. This week: the organisation that lets a company actually run them — still inside the Diagnose phase (Milestone 1).

Telefónica O2 — the programme that actually scaled

The case in brief Telefónica O2 ran a real RPA programme inside its back office. A two-process pilot went live in about two weeks and worked — the easy part. The case is about the climb that followed: taking it from two processes to running the whole back office, where most of the real work, and the real risk, live.

The setting

A major UK telecom. The work is ordinary, high-volume, rules-based back-office administration — the repetitive transactions that run an operation, and the same kind of process you flagged in your Week 4 scan.

Why we study it

An independent, peer-reviewed field study — Lacity & Willcocks (2016), open-access, not a vendor success story. One of the few rigorous accounts of an RPA programme that actually scaled.

The question on the table

A working pilot proves the logic, not the operation that must run every day. O2 made the climb; the interesting part is how. You'll make the first call yourself in a moment.

8:28

You run O2's Back Office. The pilot works at 20 bots. Do you scale?

In your team, commit to one — then we'll see what O2 did, and how lightweight RPA compared to a heavier BPMS rebuild.

A — Scale now

Push more bots onto the same setup and keep the momentum. The pilot works — why wait?

B — Fix the plant first

Re-engineer the infrastructure for load before adding bots. Slower start, fewer surprises.

C — Hand it to IT

Give it to IT to build on their queue and standards. Safe — but no longer yours to run.

Commit before you turn the page. One of these cost O2 ~16 weeks of rebuild. Decide which — then see.

The operating model was the differentiator — not the technology

Back-Office-owned

The programme lived in the Back Office, not IT — builders were reassigned process staff, so no wait on IT's queue.

The collapse (answer: B)

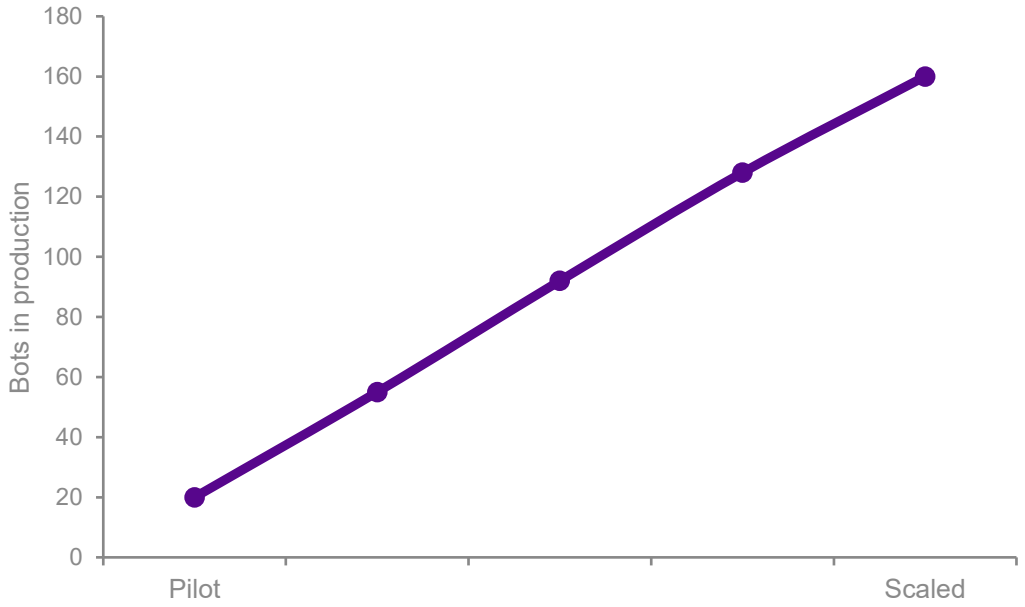
On a ‘lead’-VM setup it worked at ~20 bots; when the count quadrupled it fell over — ~16 weeks re-engineering the plant.

Same processes, two approaches: RPA (lightweight automation over existing screens) vs a heavier, IT-led BPMS. Same capability — very different payback:



“It was like driving a Ferrari with a lawn mower engine.” The gap was operating model and IT resource, not technology. (Lacity & Willcocks, 2016.)

From a two-process pilot to the back office — on three threads



~35% of back-office transactions

15 core processes · 400k–500k transactions a month

Chase-up calls down >80%

turnaround days → minutes (Customer + Cash)

~650–800% three-year ROI

hundreds of FTEs redeployed · built by 3 in-house developers

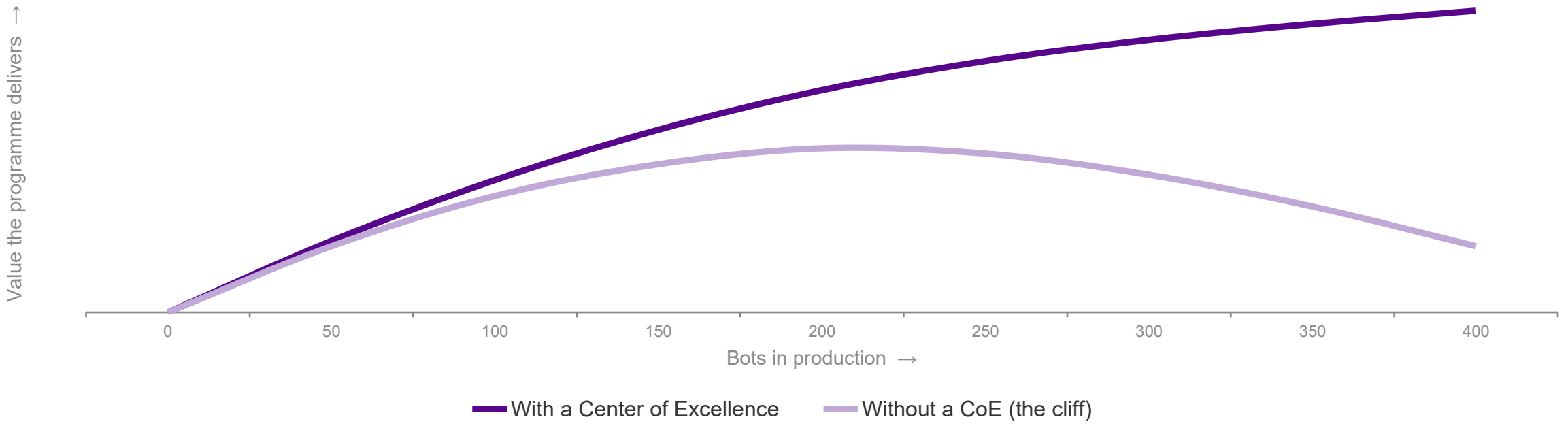
A dull capability, run well at scale. The win came from the operating model, not a clever bot — exactly the case you'll make for your own company. (Lacity & Willcocks, 2016.)

Five lessons the scaled programme leaves you

1	**A pilot proves logic, not the plant**	Working at ten bots says nothing about a hundred.
2	**Operating model beats technology**	Same tool, different owner — 10 months vs. 3 years.
3	**Own it where the work lives**	Back-Office ownership removed the IT-queue dependency.
4	**Infrastructure is a CoE job**	The lead-VM collapse is what a CoE exists to prevent.
5	**Scale the builders, not the queue**	Three builders ran 160+ bots by governing, not gatekeeping.

8:38

Ten bots is a demo; a thousand is an operation



The cliff is not a technology limit — it is a missing capability. Past a threshold, capacity goes to re-fixing broken bots instead of building. A Center of Excellence is what keeps the curve climbing — the difference between a demo and an operation.

8:40

A company has no brain for automation, so it builds one

Business

Supplies the demand and the process knowledge — what is worth automating, and how the work really runs.

→

Center of Excellence

Converts scattered requests into governed, owned, monitored automation. Governs; does not personally build everything.

→

IT

Supplies the platform, identity, and security guardrails — the plant the programme runs on.

The CoE sits between the business and IT. O2 ran all five functions with three reassigned builders; Tella's review pipeline is the sanctioned front door — a governed capability a competitor can't quickly copy, not a bigger bot factory.

Settled practice — and the hardest barriers aren't technical

74%

already running RPA — infrastructure, not an experiment.

The barriers

Process fragmentation, unclear vision, IT readiness, resistance to change — none of them the technology.

Citizen-led development

Trained non-technical staff build low-complexity automations: a complement to the CoE, under its governance — never a replacement.

The barriers are organizational — exactly what a CoE is built to answer. Programmes stall on fragmented processes, unclear vision, IT readiness, and resistance to change — people and process, not the technology.

Standardize AND empower — and the capability becomes hard to copy

Both, not either

Westerman, Ch. 3.

Standardize the platform and the guardrails; empower many people to build within them. Not a trade-off — the two together are what let automation scale.

The hard-to-copy capability

Where the advantage lives.

A rival can buy the same RPA tool tomorrow. What it can't quickly copy is the operating model that runs hundreds of governed automations — the moat is the organisation, not the software.

Scale the builders under governance — the dam with guardrails. A CoE that only builds is a bottleneck with a nicer name; one that enables supervised builders is the answer to shadow IT.

8:49

Automate the routine; route the judgement — on purpose

Attended

Runs beside a person.

Triggered by them, on their machine — a hand on the wheel, ready to take over. For work that sits near judgement.

Unattended

Runs on its own.

On a schedule or a queue, no human present — clearing routine volume overnight. For work that needs no case-by-case eyes.

A bot follows a wrong rule with perfect confidence. So route edge cases, high-value transactions, and irreversible actions to a human checkpoint. You'll name and build that oversight in Lab 2.

If a bot is silently wrong for weeks — who is accountable?

The failure

A bot follows its rule confidently even when the rule is wrong for the case in front of it — and no one notices for weeks.

The CoE answer

Bot ownership + monitoring + an audit trail: accountability is relocated to a named owner, not removed by automation.

The board question

How should liability for automated errors be structured — and what does that imply for how you design the audit?

Automation does not remove accountability; a CoE relocates it to someone who owns it. This is Board Memo §4 — and it grows louder in §7, the human-AI partnership.

Two telecoms, two kinds of evidence

Telefónica O2

Peer-reviewed (Lacity & Willcocks, 2016).

Independent research. It establishes what is true: the operating-model finding, the payback numbers, the scale. Load-bearing.

Orange Spain ‘Robot Factory’

UiPath customer case (vendor-reported).

~13 → ~100 people, 400+ bots, >€34M over a little over two years. Marketing: it illustrates a shape — it does not verify a number.

Read a vendor case critically. Never let a vendor number carry a load-bearing claim; corroborate any figure you rely on; capture title, publisher and date so a dead link stays traceable.

9:02

APPLY · BY PROJECT TEAM

Design your Center of Excellence

The next hour produces a real deliverable: the operating model a board would fund — Board Memo §4 (slide S7). Work all three asks as a team while I come round to each of you; then each team reports out, and you leave with it built.

9:05

Build your Center of Excellence — three asks, one block

Work all three in your team — I'll come round to each as you go.

1 · What breaks at scale

Take your entry-ticket risk and your §4 candidate.

Floor

– Which of the four failure modes hits your candidate first?
– Selector instability: name one system update that would break your script.
– Rank them — which breaks soonest from one bot to a hundred?

Stretch

– Which are an engineering fix, and which need an owner?
– Where would a shadow, unsanctioned version appear first?

2 · Your CoE / operating model

The deliverable — Board Memo §4.

Floor

– One line per the five functions — who does each at your company?
– Who owns, builds, governs, and funds automation?
– Scale it to your archetype (smallest may be one owner + a rule).

Stretch

– Name the hard-to-copy capability it builds (Westerman).
– How does your intake stop shadow IT before it starts?

3 · Place the human + cost it

Take one automation in your programme.

Floor

– Which edge cases, high-value, and irreversible steps get a human?
– Attended or unattended — for each bot?
– A simple ROI line: recurring work removed vs the CoE's cost.

Stretch

– If a bot is silently wrong for weeks — who's accountable, and how would an audit catch it?
– What's the workforce obligation if a role changes?

9:05–9:47 · about 14 minutes an ask. By 9:47 you have ranked scaling risks, five functions + ownership, and human checkpoints with a simple ROI line — your CoE / operating-model section, Board Memo §4 (slide S7).

9:47

Each team tells its story — together they're the class's answer

What you present

Your three answers as one line of reasoning: the risks that would break it → the operating model that holds it → where the human stays and what it costs.

The format

About 3 minutes to present, 2 minutes feedback and questions. Four teams, roughly twenty minutes — every team is heard.

What leaves the room Your CoE / operating-model section — five functions, the ownership model, the human checkpoints, and a simple ROI line. → Board Memo §4 (slide S7), still in Diagnose.

You've designed the organisation that runs automation — not another bot. Next week you build one bot in Make and break it; this CoE thinking becomes the oversight you wire in.

10:08

Next: build your first pass at home — class breaks it

Build at home, before class

Following the Lab 2 note (§3), build a first-pass Make scenario on your candidate: trigger → condition → action, connecting two systems; run it once and produce a real output. Budget ~1.5–2 hours.

Refine, break, harden — in class

Arrive with it built (or honestly attempted). Class refines the happy path, then breaks it on the flaws you seeded in Lab 1 — where it breaks is where human oversight belongs.

The gate at the door: a built (or attempted) scenario + your board deck current through S5. A refine-and-present session has nothing to refine if you never attempted the build — the deliverable is a working-workflow screenshot + a one-page post-mortem (→ §4).

10:12

Set up the connection, build the pass, and one rule

Set up first

Make + your sandbox.

Create a free Make account; set up the Connection to your Lab 1 sandbox first (import Excel to Google Sheets if needed), then connect a second system — Gmail or Slack. Half of all first-build snags are the Connection.

The one rule

Find where it breaks.

A scenario that runs perfectly taught you nothing. The failure mode you capture is the lesson — this week's fragility, met at one-bot scale.

Stuck at home is fine — empty-handed is not. If it stalls, bring the module where it broke and the exact error; the lab opens with a ~25-minute recovery window. What you can't do is arrive with nothing to refine.

10:15

Where this points

→ **§4 CoE section done** → Lab 2 (Week 6) builds the bot and the one-page post-mortem.

→ **The human-in-the-loop boundary** → carries into agent governance (§5) and the workforce question (§7).

→ **Scaling governance** → the company's Stabilise → Automate → Orchestrate roadmap, Week 11.

→ **Your judgement step** → intelligent automation (Week 7), SPAR (Week 8), the agent build (Lab 3, Week 9).

One programme, three threads. A board funds a thread outcome — Operations & Supply Chain, Cash, Customer — not a bot.

10:16

Your board deck and memo are living documents — keep them evergreen

Every week adds a slide

Each session's homework fills a defined part of the board deck and memo. Today's CoE section is Board Memo §4 / slide S7. Add it while it's fresh — don't let it queue.

Reviewed at every lab

Labs are interim reviews. Lab 2 checks two things at the door: your built scenario and your board deck current through S5. Old, unfilled slides mean you spend the lab catching up, not adding.

By December it assembles itself

Keep the deck and memo current as you go and practise the story each lab. The final board presentation is then an edit, not a scramble.

Evergreen, not end-loaded. The team that updates every week walks into the final review ready; the team that saves it for December rebuilds under pressure.

10:18

Before next week

1	**Draft**	Your CoE / operating-model section → Board Memo §4 (Automation journey, slide S7).
2	**Set up + build**	Create your Make account and build your first-pass scenario at home (Lab 2 note §3): trigger → condition → action, run once. Gated — class refines and breaks it.
3	**Bring to the lab**	A built (or honestly attempted) scenario AND your board deck current through S5 — both are checked at the door.
4	**Discussion Forum**	Forum 1: your two peer replies are due Sat Oct 3 (Brightspace → Discussions). Forum 2 — the October forum — opens later this month.

You leave with the organisation that runs automation. The operating model a board funds — not just a bot. The section is the argument; the build is Lab 2.

10:22

WEEK 5 · SCALING RPA + COE

Ten bots is a demo.
A thousand is an operation.

Next · Week 6: Lab 2 — Automate the Repeatable Part